DAY 35 - PRO

Exploratory Data Analysis & Anomaly Detection

Primary tool: Python / Statistics

What you are learning

Systematic profiling, distributions, segment comparisons and anomaly investigation.

Learning objectives

- Profile data
- Compare segments
- Flag anomalies
- Distinguish anomaly from error

Worked practical

Create an EDA report and investigate the top five anomalies.

Source / Input	Method	Output / Check
Raw data	Apply the lesson skill	Result linked to source
Change one input	Recalculate / rerun	Confirm expected change
Control total / test	Validate	Explain the result

Guided practice

1. Recreate the lesson idea on a small example.
2. Change inputs and predict the result before recalculating.
3. Complete the supplied independent dataset in the required tool.
4. Create a control check or reconciliation.
5. Explain the result in ordinary language.

AI coaching

Ask AI for plausible hypotheses, then label them explicitly as hypotheses until verified.

AI discipline

Attempt first. Use AI for explanation, hints, debugging, critique, test cases and mock questioning. Never submit output you cannot explain and independently verify.

Independent exercise

Perform structured EDA and anomaly investigation.

Common failure modes

- Wrong grain or key.
- Formula/query/code answers a different question.
- Missing or unmatched records are silently ignored.
- Findings are not reconciled to the numbers.
- AI-generated logic is accepted without testing.

Mastery gate

[ ] I can reproduce the method without copying.
[ ] I can trace the result to the source.
[ ] I have a QA/control check.
[ ] I can explain a limitation.
[ ] I can answer an assessor's follow-up question without asking AI.